Joydeep Biswas

Associate Professor, Department of Computer Science, The University of Texas at Austin, TX 78712, USA
joydeepb@cs.utexas.edu, https://www.joydeepb.com

Appointments and Education

2023 – Present	Associate Professor, Computer Science, University of Texas at Austin
2026 – Present	Director, Center for Human and Robot Co-Adaptation (NSF Science and Technology Center)
Present	Associate Director, Texas Robotics, University of Texas at Austin
2024 – Present	Visiting Professor, NVIDIA
2019 – 2023	Assistant Professor, Computer Science, University of Texas at Austin
2015 – 2019	Assistant Professor, College of Information and Computer Sciences, UMass Amherst
2014	Ph.D. in Robotics, Carnegie Mellon University
2008	B.Tech. in Engineering Physics, Indian Institute of Technology, Bombay

Teaching

Core Curriculum (University-Wide, Open to All Majors)

CS 304I: The Essentials of AI for Life and Society, UT Austin — Fall 2025, Fall 2026
CS 109: The Essentials of AI for Life and Society, UT Austin (pilot offering) — Fall 2023

- Co-designed and launched a full-scale AI literacy course with no prerequisites, accessible to students from every college, covering how AI works and its societal, ethical, and policy implications. Course design and outcomes published at EAAI 2025 and EAAI 2026 (see below).

Undergraduate Courses

CS 378H: F1/10 Autonomous Driving – Honors, UT Austin — Spring 2022, Fall 2023
CS 378F: F1/10 Autonomous Driving, UT Austin — Spring 2020, Spring 2021
CS 378/ME 379M/ECE 379K: Connected Autonomous Electric Vehicles, UT Austin, cross-listed — Spring 2023
COMPSCI 220: Programming Methodology, UMass Amherst — Fall 2017, Fall 2018
COMPSCI 403: Introduction to Robotics, UMass Amherst — Fall 2016, Spring 2018

Graduate Courses

CS 393R: Autonomous Robots, UT Austin — Fall 2020, Fall 2021, Spring 2024
CS 395T: Planning, Search, and Reasoning Under Uncertainty, UT Austin — Spring 2023
CS 388U: Planning, Search, and Reasoning Under Uncertainty, UT Austin Online MS — Fall 2023
COMPSCI 603: Robotics, UMass Amherst — Spring 2016, 2017, 2019
COMPSCI 691BR: Building a Robot Soccer Team (Seminar), UMass Amherst — Spring 2017

Teaching-Related Research

[1] Zifan Xu, Kristen Procko, Michael Munje, Kristin Patterson, Lea Sabatini, Joydeep Biswas, and Peter Stone. "The Essentials of AI for Life and Society: A Full-Scale AI Literacy Course Accessible to All." In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. 47. 2026, pp. 40730–40736.

[2] Joydeep Biswas, Don Fussell, Peter Stone, Kristin Patterson, Kristen Procko, Lea Sabatini, and Zifan Xu. "The Essentials of AI for Life and Society: An AI Literacy Course for the University Community." In: *Educational Advances in Artificial Intelligence*. Vol. 39. 28. 2025, pp. 28973–28978.

[3] Johannes Betz, Hongrui Zheng, Felix Jahncke, Zirui Zang, Florian Sauerbeck, Y. Rosa Zheng, Joydeep Biswas, Venkat Krovi, and Rahul Mangharam. "F1TENTH: Enhancing Autonomous Systems Education Through Hands-On Learning and Competition." In: *IEEE Transactions on Intelligent Vehicles* (2024).

[4] Joshua Vekhter and Joydeep Biswas. "Responsible Robotics: A Socio-Ethical Addition To Robotics Courses." In: *Educational Advances in Artificial Intelligence*. 2023, pp. 15877–15885.

[5] Joseph Spitzer, Joydeep Biswas, and Arjun Guha. "Making High-Performance Robots Safe and Easy to Use for an Introduction to Computing." In: *Educational Advances in Artificial Intelligence*. 2020, pp. 13469–13476.

Open educational resources: Interactive Particle Filters (browser-based visualization for teaching robot localization); robo-js (safe, high-performance robots for introductory computing); public course websites and assignments for all UT robotics courses.

Teaching-Related Service

- Co-PI, NSF Research Traineeship (NRT-AI): *Convergent, Responsible, and Ethical Artificial Intelligence Training Experience for Roboticists*, 2021 – 2026
- Student Chair, Association for the Advancement of Artificial Intelligence (AAAI): current
- Program Committee, AAAI Symposium on Educational Advances in AI and AAAI Undergraduate Consortium: 2021
- UTCS Turing Scholars (Honors) Admissions Committee: 2020 – 2022
- UMass CICS Honors Program Director; Undergraduate Course Assistant Program Director: 2018 – 2019
- Co-Organizer, Texas Regional Robotics Symposium: 2022
- Outreach: *HolyokeCodes* week-long robotics workshop for high-school students (2019); public science talks at SciTech Café (2017) and Amherst Cinema *Science on Screen* (2018); UT Good Systems live autonomous delivery webcast (2020)

Advising

Postdoctoral Scholars

Rohan Chandra (2022 – 2024, now Assistant Professor, University of Virginia); Kiarash Rahmani (2022 – 2024, now Applied AI Scientist, Durable AI); Nathan Tsoi (2025 – present)

PhD Students

Graduated: Amanda Adkins (2026, now at Amazon); Sadegh Rabiee (2022, now at Amazon Lab126); Jarrett Holtz (2022, now at Bosch Research)
Current: Taijing Chen, Sarah Etter, Dongmyeong Lee, Arthur Zhang, Zichao Hu, Sadanand Modak, Eric Hsiung

Master's Thesis Students

Kavan Sikand (UT Austin, 2022); David Balaban (UMass Amherst, 2018)

Undergraduate Honors Thesis Students

- **UT Austin (2021 – 2022):** Elvin Yang (**Best Honors Thesis Award**), Rahul Menon, Shakeel Samsudeen, Nathaniel Plaxton, Michael Satanovski
- **UMass Amherst (2015 – 2019):** Edward Schneeweiss, Kyle Vedder, George Larionov

Committees: PhD committees: 9 (UT Austin and UMass Amherst); external thesis opponent, KTH Royal Institute of Technology; 2 undergraduate honors thesis committees

Teaching and Mentoring Awards/Honors

2025, 2026	CRA Outstanding Undergraduate Researcher Award: **Awardee** (2026) and Honorable Mention (2025), advisee Luisa Mao
2024	CRA Outstanding Undergraduate Researcher Award, Honorable Mention (advisee Jimmy Xin)
2022	Best Honors Thesis Award, UT Austin Computer Science (advisee Elvin Yang)
2020	CRA Outstanding Undergraduate Researcher Award, Honorable Mention (summer research advisee Jenna Hammond, Mount Holyoke College)
2019	Student Best Poster Award, Northrop Grumman University Symposium (advised students)
2015 – 2018	RoboCup Small Size League: 1st place 2015 (*CMDragons*, student team leader); Lower Bracket 1st place 2017 (*UMass MinuteBots*, faculty team leader of student team)

Selected Research Honors

Chief of Naval Research Fellow, inaugural class (2026); NSF CAREER Award (2021); AAAI Senior Member (2026); IEEE Senior Member (2026); Best Paper Award, ACM CHI (2024); Best Paper Honourable Mention, HRI (2026); Best Paper Award Finalist, ICRA (2026); IJCAI Early Career Spotlight (2019); Amazon Research Award (2018); JP Morgan Faculty Research Awards (2018, 2024); Siebel Scholar (2015)